

Identifying and Forecasting US Equity Market Regimes: An Unsupervised and Supervised Machine Learning Analysis of Bull and Bear Market States, 2010–2025

A Research Paper on Quantitative Finance and Market Regime Classification

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Abstract—This paper investigates whether unsupervised machine learning can identify economically meaningful regimes in the US equity market and whether such regimes provide genuine forecasting or investment value. Using a daily dataset of 3,823 observations spanning 18 October 2010 to 31 December 2025, sixteen standardized market, volatility, and macro-financial features are reduced via Principal Component Analysis (eight components explaining 92.1% of variance) and partitioned using K-Means clustering, validated against a Gaussian Mixture Model. The optimal solution, selected by silhouette score, identifies two statistically and economically distinct regimes, subsequently labelled Bull / Low-Risk ($n = 3,201$; 83.7% of observations) and Bear / High-Risk ($n = 622$; 16.3%). The regimes differ significantly across mean return, volatility, VIX level, momentum, and drawdown (Welch t-tests, all $p < .001$; Cohen's d up to 2.94), and exhibit strong persistence, with Bull-to-Bull and Bear-to-Bear transition probabilities of 96.5% and 81.8% respectively. A chronological, one-step-ahead out-of-sample forecasting exercise compares Logistic Regression, Random Forest, and Gradient Boosting classifiers against a naive persistence benchmark (“tomorrow's regime equals today's regime”). The persistence benchmark achieved the highest out-of-sample accuracy (93.55%), exceeding Random Forest (91.89%), Logistic Regression (89.80%), and Gradient Boosting (89.01%). This finding does not support the hypothesis that supervised machine learning provides superior next-day regime forecasting relative to a simple persistence rule. A regime-conditioned tactical investment strategy (fully invested in Bull, fully divested in Bear, signal lagged by one day, 10 basis point transaction costs) reduced maximum drawdown relative to buy-and-hold (−28.20% versus −33.92%) but underperformed on both compound annual growth rate (5.80% versus 12.68%) and Sharpe ratio (0.546 versus 0.780). Forward-return analysis indicates that returns following Bear/High-Risk observations were, on average, higher than returns following Bull/Low-Risk observations at 1-, 5-, and 20-day horizons, offering a plausible explanation for the strategy's return shortfall despite its lower drawdown. The study concludes that regime identification is statistically robust and economically interpretable, but that regime identification is not equivalent to reliable regime forecasting or to superior risk-adjusted investment performance. The paper's contribution lies in this explicit

separation of unsupervised classification, supervised forecasting, and economic usefulness — three distinct questions that are frequently conflated in applied machine learning finance research. A supplementary robustness check (Section XI.J), which widens the feature set to twenty-nine variables by restoring the raw price, volume, and moving-average series the main specification deliberately excludes, produces a weaker four-regime solution and a severe train/test distribution shift that collapses out-of-sample forecasting performance — a negative result that, if anything, reinforces the original feature-exclusion rationale in Section IV.

Keywords—*Market Regimes; K-Means Clustering; Gaussian Mixture Models; Principal Component Analysis; US Equity Market; Volatility Regimes; Regime Forecasting; Machine Learning in Finance; Persistence Benchmark; Tactical Asset Allocation*

I. INTRODUCTION

A. Background

Financial markets do not generate returns from a single, time-invariant statistical process. Return volatility clusters, correlations between assets shift, and the relationship between risk and reward appears to differ systematically across periods commonly described as “bull” and “bear” markets ([8], [10]). This observation has motivated a long tradition of regime-based modelling in financial econometrics, from Markov-switching models of the business cycle ([10]) to regime-dependent asset allocation frameworks ([1], [2]). More recently, the availability of large panels of market, volatility, and macro-financial indicators has encouraged researchers to apply unsupervised machine learning methods — clustering algorithms in particular — as a data-driven alternative to parametric regime-switching models ([15]).

B. Research Problem

Identifying whether the market is currently in a high-risk, drawdown-prone state or a low-risk, trending state matters both for academic characterisation of return dynamics and for practical risk management. If regimes can be identified reliably and, more importantly, forecast before they fully unfold, investors could in principle adjust exposure to reduce downside risk without sacrificing long-run returns. The practical value of any regime framework therefore rests on two separate capabilities: the ability to describe the past accurately, and the ability to anticipate the future usefully.

C. Gap

A recurring problem in this literature is that the two capabilities keep getting conflated. It is common to see a paper report high in-sample or same-day classification accuracy for a supervised model trained to recover a regime label that was itself built from the same contemporaneous features the model is then “predicting” with. That result is close to circular — it shows a classifier can recover its own clustering inputs, not that it can forecast a genuinely future, economically unknown state. A second, related limitation shows up in studies built around a single regime or a single market cycle: they tend to quietly assume a stable, unchanging data-generating process, when in practice the frequency, duration, and severity of regimes appear to shift substantially over time ([16]). This paper tries to avoid the first problem by keeping same-day regime characterisation strictly separate from genuine one-step-ahead forecasting, and addresses the second by reporting regime duration and persistence statistics directly from the data rather than assuming stability.

D. Research Objective

The objective of this study is threefold: (i) to identify statistically and economically distinct market regimes in US equity data using unsupervised machine learning; (ii) to test, using a strictly chronological out-of-sample design, whether supervised machine learning models can forecast next-day regime membership better than a naive persistence benchmark; and (iii) to evaluate whether the identified regimes have practical investment value once translated into a simple, cost-adjusted tactical allocation strategy.

E. Research Questions

RQ1: Can US equity market observations be partitioned into statistically and economically distinct regimes using unsupervised machine learning?

RQ2: Do the identified regimes exhibit meaningful temporal persistence, and how long do they typically last?

RQ3: Can supervised machine learning models forecast next-day regime membership more accurately than a naive persistence rule?

RQ4: Does a regime-conditioned tactical investment strategy improve risk-adjusted returns relative to a buy-and-hold benchmark?

F. Contributions

This study contributes to the literature in three ways. First, it applies a complete, reproducible unsupervised pipeline (standardisation, PCA, K-Means, and Gaussian Mixture Model validation) to a sixteen-year daily US equity dataset and reports the full set of diagnostic statistics necessary to assess cluster validity. Second, it implements a forecasting design that is explicitly non-circular: predictors are lagged and the target is the following day’s regime, with a naive persistence baseline reported alongside every supervised model so that any claimed forecasting skill can be benchmarked rather than asserted. Third, it connects statistical regime identification to an implementable, cost-adjusted investment strategy and reports the realised trade-off between drawdown reduction and return, rather than presenting classification accuracy as a proxy for investment usefulness.

G. Paper Structure

Section II reviews the relevant literature on market regimes, clustering methods, and machine learning in finance. Section III states the research questions and hypotheses. Section IV describes the dataset and variables. Section V details the methodology. Sections VI through IX present empirical results covering regime identification, forecasting performance, feature importance, and tactical investment strategy evaluation.

Section X discusses the findings, Section XI addresses robustness and limitations, and Section XII concludes.

II. LITERATURE REVIEW

A. Financial Market Regimes and Bull/Bear Markets

The idea that markets alternate between qualitatively distinct states has a long history in financial economics. Fama [8] provided the foundation against which subsequent regime-based critiques were developed, and [16] proposed one of the more influential rule-based algorithms for dating bull and bear markets from price data alone, using drawdown and duration filters conceptually related to the drawdown- and momentum-based features used in the present study.

B. Markov-Switching and Regime-Dependent Models

[10] introduced the regime-switching autoregressive framework that remains the dominant parametric approach to modelling discrete shifts in the mean and variance of macroeconomic and financial time series. [1] extended regime-switching methodology to international asset allocation, showing that correlations and volatilities rise in “bad” regimes and that ignoring this structure understates portfolio risk. [2] provide a comprehensive review of regime-change models in finance, noting that regimes identified in financial data are typically persistent — a finding directly consistent with the transition probabilities reported in Section IX of this paper.

C. Volatility Regimes and the VIX

The CBOE Volatility Index (VIX) is widely used as a forward-looking gauge of market-implied risk and is closely associated with regime shifts ([18]). [5] documents a set of “stylised facts” of asset returns, including volatility clustering and leverage effects, that are consistent with the pronounced differences in volatility and VIX level observed between the two regimes identified in this study.

D. Unsupervised Machine Learning in Finance: K-Means and Gaussian Mixture Models

K-Means clustering ([14]) partitions observations to minimise within-cluster variance and has become a common tool for data-driven regime identification because it requires no prior distributional assumptions about regime membership. Gaussian Mixture Models extend this idea by allowing soft, probabilistic cluster assignment under a mixture-of-Gaussians assumption ([17]), and are frequently used as a robustness check against the harder partitions produced by K-Means, as is done in this paper.

E. Principal Component Analysis in Financial Markets

Principal Component Analysis ([12]) is a standard dimensionality-reduction technique used to address multicollinearity among financial indicators and to concentrate the signal in a small number of orthogonal components prior to clustering. In the present dataset, PCA is used specifically to

prevent variables with different natural scales — for example, the VIX level relative to daily percentage returns — from disproportionately driving the clustering solution.

F. Machine Learning for Financial Prediction and Its Limitations

[9] provide a large-scale comparative study of machine learning methods for cross-sectional return prediction, finding that tree-based and neural network methods can improve on linear benchmarks but that gains are modest relative to the noise inherent in financial returns. [3] introduced the Random Forest algorithm used in this study, and [11] provide the standard reference for the logistic regression and ensemble methods applied here. A recurring caution in this literature is that classification accuracy alone can be a misleading indicator of forecasting skill when the target class is highly imbalanced or persistent, precisely the situation this paper addresses directly through its persistence baseline.

G. Regime-Based and Tactical Asset Allocation

[7] popularised simple moving-average-based tactical allocation rules as a means of reducing drawdown, and subsequent work, including [15], has examined dynamic portfolio optimisation across machine-learning-identified hidden regimes. A consistent theme in this literature is that regime-based or trend-following strategies tend to reduce maximum drawdown relative to buy-and-hold but do not uniformly improve compound returns, particularly once transaction costs and the opportunity cost of exiting the market during volatile-but-ultimately-recovering periods are taken into account — a trade-off that is examined empirically for the present dataset in Section X.

III. RESEARCH QUESTIONS AND HYPOTHESES

The following hypotheses translate the research questions in Section I.E into testable statements. Each hypothesis is evaluated strictly against the Replication Data Archive; where a hypothesis is not supported, this is stated explicitly rather than reframed post hoc.

H1: US equity market observations can be separated into statistically and economically distinct market regimes using unsupervised clustering.

H2: The Bear/High-Risk regime exhibits significantly lower returns and significantly higher volatility, VIX level, and drawdown than the Bull/Low-Risk regime.

H3: The identified market regimes exhibit meaningful temporal persistence rather than switching randomly from day to day.

H4: Supervised machine learning models provide superior next-period regime forecasting accuracy compared with a naive persistence benchmark.

As reported in Sections VI and VII, H1, H2, and H3 are supported by the evidence. H4 is not supported: the persistence benchmark outperformed every supervised model tested. This result is retained and discussed rather than revised, consistent with the paper's stated commitment to reporting findings as observed.

IV. DATA AND VARIABLES

The dataset consists of 3,823 daily observations of a US equity benchmark (S&P 500 level) and associated market, volatility, and macro-financial indicators, spanning 18 October 2010 to 31 December 2025. No missing values were present in any of the thirty-three columns of the cleaned dataset (Table 1 reports the underlying data-quality check). The S&P 500 closing level ranges from approximately 1,099 to 6,932 over the sample, and the VIX ranges from 9.14 to a maximum of 82.69, consistent with the inclusion of a major volatility event (the 2020 COVID-19 market shock) within the sample period.

Sixteen candidate features were used for regime discovery: Return, Log_Return, Volatility_20D, Volatility_60D, Momentum_20D, Momentum_60D, Distance_MA50, Distance_MA200, Drawdown, Volume_Ratio, VIX, VIX_Change, VIX_Ratio, Treasury_20D_Change, and Oil_Return. Raw OHLC price levels, raw trading volume, and the running maximum price were deliberately excluded from the clustering feature set because they are non-stationary and would dominate the clustering solution through scale and price-level drift rather than through genuine regime information.

Table 1. Clustering variables used for regime discovery and their economic interpretation.

Variable	Definition	Economic Interpretation
Return	Daily percentage change in the S&P 500 closing level	Contemporaneous market performance
Log_Return	Natural-log daily return	Return measure suitable for compounding/statistical tests
Volatility_20D / 60D	Rolling 20-/60-day standard deviation of returns, annualised	Realised risk over short and medium horizons
Momentum_20D / 60D	Cumulative return over the trailing 20/60 trading days	Trend strength and direction
Distance_MA50 / MA200	Percentage distance of price from its 50-/200-day moving average	Position relative to short- and long-term trend
Drawdown	Percentage decline from the running maximum price	Depth of current loss from peak
Volume_Ratio	Trading volume relative to its 20-day moving average	Relative market participation/liquidity

Variable	Definition	Economic Interpretation
VIX	CBOE Volatility Index level	Market-implied (forward-looking) risk expectation
VIX_Change / VIX_Ratio	Day-over-day change / ratio of VIX to its 20-day average	Short-term shift in risk sentiment
Treasury_20D_Change	20-day change in the 10-year US Treasury yield	Macro/interest-rate environment
Oil_Return	Daily return on WTI crude oil	Commodity/macro risk co-movement

Note: All variables were standardised (zero mean, unit variance) prior to PCA and clustering; the scaler was fit on the training period only to avoid look-ahead bias. Raw OHLC prices, raw volume, and Running_Max were excluded from the clustering feature set for the reasons stated in Section IV.

V. METHODOLOGY

A. Data Preprocessing

Infinite values arising from ratio-based feature construction were replaced with missing values and dropped, and duplicate dates were removed, leaving 3,823 clean daily observations with zero missing values across all analysis variables (Table 1, Data Quality panel). The data were sorted chronologically and a 70/30 chronological split was used throughout — the first 70% of observations as the training period and the final 30% as the out-of-sample test period — rather than a random train/test split, because random shuffling of time-ordered financial data allows information from the future to leak into the training set and produces artificially inflated performance estimates.

B. Standardisation

Because the clustering variables are measured on very different natural scales (for example, the VIX ranges roughly from 9 to 83, while daily returns are typically within $\pm 10\%$), each feature x was standardised as $z = (x - \mu) / \sigma$, where μ and σ are the mean and standard deviation estimated on the training period only. The fitted scaler was then applied, without re-estimation, to the test period and to the full sample, preventing information from the test period from influencing the standardisation statistics.

C. Principal Component Analysis

PCA was applied to the standardised training features to obtain an orthogonal set of components ordered by explained variance. The number of components was chosen as the smallest number for which cumulative explained variance reached at least 90%. In this dataset, eight principal components were required, jointly explaining 92.1% of total variance (Table 2). The first principal component alone explains 36.7% of variance and loads positively on Drawdown (0.379), Distance_MA50 (0.347), Distance_MA200 (0.341), Momentum_60D (0.311), and Momentum_20D (0.296), and negatively on VIX (−0.338) and Volatility_20D (−0.325); it is therefore readily interpretable as a general trend/risk axis, with high values corresponding to uptrending, low-volatility conditions and low values corresponding to drawdown-heavy, high-volatility conditions. The second component (20.0% of variance) loads most heavily on same-day Return and Log_Return and on VIX_Change, and is better interpreted as a short-horizon return/volatility-shock axis rather than a persistent-state axis.

D. K-Means Clustering

K-Means clustering partitions observations into K clusters by minimising the within-cluster sum of squared Euclidean distances to each cluster centroid. The number of clusters K was selected by fitting K-Means for $K = 2$ through 6 on the training-period principal component scores and comparing silhouette

scores, which measure how well each observation fits within its assigned cluster relative to the nearest neighbouring cluster. The silhouette score was maximised at $K = 2$ (0.466), declining monotonically for $K = 3$ (0.410) through $K = 5$ (0.156) before a marginal uptick at $K = 6$ (0.166) that did not exceed the $K = 2$ solution (Table 3). $K = 2$ was therefore selected as the optimal number of regimes, and the final K-Means model was refit with 50 random initialisations for stability.

E. Gaussian Mixture Model

As a robustness check on the hard $K = 2$ partition produced by K-Means, a Gaussian Mixture Model with two full-covariance components was independently fit to the same principal component scores. GMM assigns each observation a posterior probability of belonging to each component rather than a single hard label, which allows the plausibility of a genuinely bimodal (rather than K-Means-imposed) structure in the data to be assessed. Raw label agreement between the two methods was computed as a diagnostic; because cluster labels are arbitrary numerical identifiers rather than ordered categories, this raw agreement rate is reported as a descriptive diagnostic only and is not treated as a definitive similarity measure.

F. Regime Identification and Labelling

Cluster labels produced by K-Means are arbitrary integers (0, 1) with no inherent economic meaning. To assign an interpretable label, the mean daily return of each cluster was computed, and the cluster with the higher mean return was labelled “Bull / Low-Risk” while the cluster with the lower mean return was labelled “Bear / High-Risk.” This labelling is descriptive and ex-post: it characterises what each numerically identified cluster looks like once fully formed, using information (contemporaneous mean return) that would not be available in real time on any single day. This distinction between ex-post descriptive labelling and real-time regime detection is maintained throughout the paper and is revisited explicitly in Section XI.

G. Statistical Validation

Differences between the two regimes were tested using the Welch two-sample t-test (which does not assume equal variances, appropriate given the large volatility difference between regimes), the non-parametric Mann-Whitney U test (which does not assume normally distributed returns), and Cohen's d as a standardised effect-size measure. A non-parametric bootstrap with 5,000 resamples was additionally used to construct a 95% confidence interval around the observed difference in mean daily return between regimes, providing a distribution-free check on the parametric test results.

H. Regime Transition Analysis

A first-order transition matrix was constructed by cross-tabulating each day's regime against the following day's regime and normalising by row, giving the empirical probability of remaining in, or switching out of, each regime. Regime episodes (contiguous runs of the same regime label) were separately identified and their duration in trading days and calendar days was summarised to characterise typical regime length.

I. Supervised Machine Learning Forecasting

To avoid the circularity of predicting a regime label using the same contemporaneous features that generated it, the supervised forecasting exercise predicts tomorrow's regime using today's (lagged) features. Three classifiers — Logistic Regression, Random Forest, and Gradient Boosting — were trained on the chronological first-70% training period and evaluated on the final 30% test period, with performance compared against a naive persistence baseline that simply predicts tomorrow's regime to equal today's regime. Accuracy, precision, recall, and F1-score were computed for each model on the held-out test period. Because regimes are persistent by construction (Section IX), the persistence baseline is expected to be a demanding benchmark; a supervised model is only demonstrating genuine forecasting skill to the extent that it exceeds this baseline, not merely to the extent that its raw accuracy is high.

J. Investment Strategy Construction

A simple regime-conditioned tactical strategy was constructed: the strategy holds a 100% equity position on days following a Bull/Low-Risk regime signal and a 0% position (fully in cash) on days following a Bear/High-Risk regime signal. The regime signal is deliberately lagged by one trading day (today's exposure is set using yesterday's realised regime) to avoid the look-ahead bias of using a day's own regime label to explain that same day's return. A transaction cost of 10 basis points (0.10%) is applied to every unit change in position (i.e., every full transition between fully invested and fully divested). Strategy and buy-and-hold equity curves were constructed by compounding daily net returns, and performance was summarised using total return, compound annual growth rate (CAGR), annualised volatility, Sharpe ratio, maximum drawdown, and the Calmar ratio (CAGR divided by the absolute value of maximum drawdown).

VI. EMPIRICAL RESULTS

A. PCA Results

Table 2 reports the explained variance of each of the sixteen principal components estimated on the training period. The first two components jointly account for 56.6% of total variance, and the eight-component solution used for clustering accounts for 92.1%. This confirms that the underlying feature space, while sixteen-dimensional, is effectively driven by a smaller number of latent factors, the most important of which (PC1) closely resembles a composite trend/risk axis combining trend-following variables (moving-average distance, momentum, drawdown) with contemporaneous risk variables (volatility, VIX) in opposing directions.

Table 2. PCA explained variance (training period), first eight components (retained for clustering).

Component	Explained Variance	Cumulative Variance
PC1	36.67%	36.67%
PC2	19.97%	56.64%
PC3	9.62%	66.26%
PC4	6.84%	73.11%
PC5	5.64%	78.74%
PC6	5.36%	84.10%
PC7	4.57%	88.67%
PC8	3.47%	92.14%

Note: Components 9–16 (not shown) contribute the remaining 7.9% of variance in progressively diminishing amounts; the full sixteen-component table is available in the Supplementary Appendix (Supplementary Appendix, Table S2).

B. Cluster and Regime Results

The silhouette-score comparison across candidate values of K (Table 3) identifies $K = 2$ as the optimal number of clusters, decisively outperforming $K = 3$ through $K = 6$. The resulting two-regime solution, labelled Bull/Low-Risk and Bear/High-Risk according to the procedure in Section V.F, produces the descriptive statistics in Table 4.

Table 3. K-Means cluster selection diagnostics, K = 2 through 6 (training-period PCA scores).

K	Inertia	Silhouette Score
2	30,204.67	0.466
3	26,779.36	0.410
4	24,144.62	0.270
5	22,565.16	0.156
6	21,129.19	0.166

Table 4. Descriptive statistics by identified regime, full sample.

Metric	Bull / Low-Risk	Bear / High-Risk
Observations	3,201 (83.7%)	622 (16.3%)
Mean daily return	+0.1376%	−0.3810%
Median daily return	+0.1017%	−0.4117%
Std. dev. of daily return	0.7387%	2.0466%
Annualised volatility	11.73%	32.49%
Mean VIX	16.16	28.30
Mean drawdown	−2.42%	−13.12%
Mean 20-day momentum	+1.91%	−3.62%
Mean 60-day momentum	+4.90%	−6.22%

Note: Annualisation of return and volatility uses 252 trading days. Values computed directly from the classified dataset; see Replication Data Archive, Table S4.

C. Economic Interpretation

The two regimes are economically coherent with the standard bull/bear characterisation. The Bull/Low-Risk regime displays a positive mean daily return, low realised volatility (11.7% annualised), a low average VIX (16.2, close to its long-run historical norm), positive momentum at both the 20- and 60-day horizons, and a shallow average drawdown from peak (−2.4%). The Bear/High-Risk regime displays a negative mean daily return, volatility nearly triple that of the Bull regime (32.5% annualised), an elevated average VIX (28.3, consistent with heightened risk aversion), negative momentum, and a materially deeper average drawdown (−13.1%). This pattern across five independent economic dimensions — return, volatility, implied risk, trend, and drawdown — supports the

interpretation of the clusters as genuine risk-return regimes rather than statistical artefacts of the clustering procedure.

Fig. 1. S&P 500 price level coloured by identified market regime (Bull/Low-Risk vs Bear/High-Risk), 2010–2025.

This figure plots the S&P 500 closing level over the full sample, with each observation coloured according to its classified regime. Bear/High-Risk observations cluster visibly around the major drawdown episodes in the sample (notably the sharp 2020 decline), consistent with the regime's economic characterisation in Section VI.C.

Fig. 2. Distribution of daily returns by regime.

The Bear/High-Risk return distribution is visibly wider and left-shifted relative to the Bull/Low-Risk distribution, consistent with the higher volatility and negative mean return reported in Table 4.

Fig. 3. VIX distribution across regimes (boxplot).

The VIX distribution for the Bear/High-Risk regime is shifted markedly higher, with a wider interquartile range, than the Bull/Low-Risk regime, corroborating the mean VIX difference (16.16 vs 28.30) reported in Table 4.

D. Statistical Significance

Table 5 reports Welch t-tests, Mann-Whitney U tests, and Cohen's d effect sizes for the difference between regimes across eight variables. Every comparison is statistically significant at conventional levels, with p-values far below .001 in all cases; given the large sample size ($n = 3,823$), however, statistical significance alone is not informative, and the effect sizes are the more economically meaningful figures. Cohen's d exceeds 2.0 in magnitude for VIX (−2.37), 20-day volatility (−2.08), drawdown (2.94), 60-day momentum (2.26), and distance from the 200-day moving average (2.85) — conventionally “huge” effect sizes, well beyond the $d = 0.8$ threshold normally considered a large effect ([4]). Daily return itself shows a more moderate, though still statistically significant, effect ($d = 0.49$).

Table 5. Welch t-test and Cohen's d effect sizes, Bull/Low-Risk vs Bear/High-Risk regime.

Variable	Bull Mean	Bear Mean	Welch t	p-value	Cohen's d
Daily Return	0.1376 %	-0.3810 %	6.24	<.001	0.486
VIX	16.16	28.30	-33.66	<.001	-2.368
20D Volatility	0.1205	0.2753	-25.60	<.001	-2.079
60D Volatility	0.1329	0.2499	-23.23	<.001	-1.663
Drawdown	-2.42 %	-13.12 %	46.90	<.001	2.936
20D Momentum	1.91%	-3.62 %	21.81	<.001	1.555
60D Momentum	4.90%	-6.22 %	41.23	<.001	2.260
Distance from MA200	6.65%	-5.00 %	52.91	<.001	2.850

Note: Mann-Whitney U statistics (unreported in this table for brevity) were significant at $p < .001$ for all eight variables, corroborating the parametric results without relying on a normality assumption. Full statistics are provided in Replication Data Archive, Table S6.

An important caveat applies to the interpretation of these effect sizes: because several of the tested variables (Return, VIX, Volatility_20D, Drawdown) were also among the sixteen features used to construct the underlying clusters via PCA, part of the separation reported here is a near-mechanical consequence of the clustering procedure rather than fully independent, out-of-sample validation. This is standard practice in cluster-characterisation studies and does not invalidate the regime interpretation, but it does mean the effect sizes should be read as confirming that the clusters are coherent along their own construction dimensions, rather than as an independent test of predictive validity.

E. Bootstrap Analysis

A non-parametric bootstrap (5,000 resamples) of the difference in mean daily return between the Bull and Bear regimes produced a 95% confidence interval of [0.359%, 0.677%] around the observed difference of 0.519%. Because this interval excludes zero by a wide margin, the bootstrap result corroborates the parametric and non-parametric hypothesis tests reported in Section VI.D using a distribution-free method that does not rely on any assumption about the shape of the return distribution.

F. Regime Persistence

Table 6 reports the first-order transition matrix. The Bull/Low-Risk regime is highly persistent: conditional on being in the Bull regime on a given day, there is a 96.47% probability of remaining in the Bull regime the following day, versus only a 3.53% probability of transitioning to Bear. The Bear/High-Risk regime is somewhat less persistent but still shows strong self-continuation, with an 81.83% probability of remaining in Bear versus an 18.17% probability of transitioning back to Bull.

Fig. 4. Regime transition probability matrix (heatmap).

This heatmap visualises the transition probabilities reported in Table 6, highlighting the strong diagonal persistence of both regimes, particularly the Bull/Low-Risk regime.

Table 6. First-order regime transition probability matrix.

Current Regime \ Next Regime	Bull / Low-Risk	Bear / High-Risk
Bull / Low-Risk	96.47%	3.53%
Bear / High-Risk	18.17%	81.83%

Regime-episode analysis (Table 7) shows that the two regimes differ markedly in their duration distribution. Bear/High-Risk episodes are frequent (113 separate episodes over the sample) but typically brief, with a mean duration of 5.5 trading days and a median of just 1 trading day, alongside a small number of much longer episodes (maximum 67 trading days), indicating a highly right-skewed duration distribution dominated by short, sharp volatility spikes. Bull/Low-Risk episodes are less numerous (114 episodes) but on average much longer (mean 28.1 trading days), and here too the distribution is strongly right-skewed — the median Bull episode lasts only 3 trading days, while the longest single Bull episode extends to 404 trading days, indicating that the sample-level Bull-regime persistence statistics are disproportionately shaped by a small number of very long uptrends rather than being representative of a “typical” bull episode.

Table 7. Regime episode duration statistics.

Regime	Episodes	Mean (Trading Days)	Median (Trading Days)	Min (Trading Days)	Max (Trading Days)
Bear / High-Risk	113	5.50	1	1	67
Bull / Low-Risk	114	28.08	3	1	404

VII. MACHINE LEARNING FORECASTING RESULTS

This section reports the one-step-ahead, chronologically out-of-sample forecasting exercise described in Section V.I. All models predict tomorrow's regime using today's lagged feature values; no model has access to same-day information about the target it is predicting.

Table 8. Out-of-sample one-step-ahead regime forecasting performance (chronological 70/30 split).

Model	Accuracy	Precision	Recall	F1
Persistence Baseline	93.55%	95.79%	95.79%	95.79%
Logistic Regression	89.80%	89.27%	98.52%	93.67%
Random Forest	91.89%	92.07%	97.84%	94.86%
Gradient Boosting	89.01%	89.00%	97.72%	93.16%

Note: Persistence Baseline predicts that tomorrow's regime equals today's regime; it is not a fitted classifier but is evaluated identically to the supervised models. Source: Replication Data Archive, Table S8.

The central empirical finding of this section is that the naive persistence baseline achieved the highest out-of-sample accuracy (93.55%) of any model tested, ahead of Random Forest (91.89%), Logistic Regression (89.80%), and Gradient Boosting (89.01%). This result directly rejects Hypothesis H4: supervised machine learning did not provide superior next-period forecasting accuracy relative to a simple persistence rule in this dataset.

Fig. 5. Out-of-sample regime prediction accuracy by model, with the persistence baseline shown as a reference line.

All three supervised models fall below the persistence baseline reference line, visually confirming the result in Table 8 and the rejection of Hypothesis H4.

This outcome is not a coding artefact but a direct and expected consequence of the persistence statistics documented in Section VI.F. Because the Bull regime alone persists into the following day with 96.5% probability, and the sample is itself heavily imbalanced (83.7% Bull observations), a rule that simply repeats yesterday's label achieves high accuracy almost mechanically. A supervised classifier can only add value by correctly anticipating the comparatively rare transition days,

and Table 8 indicates that none of the three models tested were able to do so more accurately overall than the baseline. This is an important and often under-reported feature of regime-forecasting studies: high raw accuracy is not sufficient evidence of forecasting skill when the target class is persistent and imbalanced, and any claim of predictive superiority must be benchmarked against a naive persistence rule rather than judged against zero or against random-guess accuracy.

VIII. FEATURE IMPORTANCE

Although the Random Forest classifier did not outperform the persistence baseline (Section VII), its internal feature-importance ranking is informative about which lagged variables the model found most useful in attempting to distinguish forthcoming regime membership.

Table 9. Random Forest feature importance, out-of-sample forecasting model.

Feature	Importance
Distance_MA200	20.27%
Momentum_60D	18.44%
Drawdown	15.97%
Distance_MA50	13.18%
Volatility_20D	7.12%
VIX	5.97%
Momentum_20D	5.81%
Volatility_60D	3.92%
Treasury_20D_Change	2.37%
VIX_Ratio	1.78%
Return	1.53%
Oil_Return	1.42%
VIX_Change	1.13%
Volume_Ratio	1.11%

The four most important features — Distance_MA200, Momentum_60D, Drawdown, and Distance_MA50 — jointly account for 67.9% of total feature importance and are all medium-to-long-horizon trend and drawdown indicators rather than short-horizon return or volume measures. This is economically intuitive: a market's position relative to its long-run trend and the depth of its current drawdown are slow-moving state variables that plausibly carry more information about which regime is likely to persist than a single day's return or trading volume. VIX and VIX-derived variables (VIX, VIX_Ratio, VIX_Change) collectively account for a further 8.9% of importance, while macro variables (Treasury_20D_Change, Oil_Return) contribute modestly (3.8% combined). It should be stated explicitly that feature importance indicates predictive contribution within the fitted model, not causal influence; the ranking describes which lagged variables the algorithm relied upon most heavily, not which variables cause regime transitions.

Fig. 6. Random Forest feature importance for the out-of-sample regime forecasting model.

Distance from the 200-day moving average, 60-day momentum, and drawdown dominate the importance ranking, as discussed in Section VIII.

IX. INVESTMENT STRATEGY RESULTS

The regime-conditioned tactical strategy described in Section V.J was backtested over the full sample and compared against a passive buy-and-hold benchmark. Both series are computed on the same 3,823 trading days and reflect the same underlying asset.

Table 10. Regime strategy versus buy-and-hold performance, full sample (10bp transaction cost per position change).

Metric	Regime Strategy	Buy & Hold S&P 500
Total Return	135.29%	511.36%
CAGR	5.80%	12.68%
Annualised Volatility	11.56%	17.20%
Sharpe Ratio	0.546	0.780
Maximum Drawdown	−28.20%	−33.92%
Calmar Ratio	0.206	0.374
% Positive Days	45.38%	54.67%

The strategy was invested 83.7% of trading days on average and made 227 position changes over the sample, with cumulative transaction costs equal to 22.7% of one-times notional exposure summed across all trades — a material cost drag over the full 15-year horizon that is already reflected in the net performance figures above.

Fig. 7. Regime strategy vs buy-and-hold equity curves (growth of \$1).

The buy-and-hold curve ends at a materially higher terminal value than the regime strategy curve, consistent with the CAGR gap reported in Table 10, though the strategy curve shows visibly shallower dips during major drawdown periods.

Fig. 8. Drawdown comparison, regime strategy vs buy-and-hold.

The regime strategy's drawdown line stays closer to zero than the benchmark's during the deepest drawdown episodes, consistent with its lower maximum drawdown (−28.20% vs −33.92%) in Table 10.

Fig. 9. Frequency of identified market regimes (% of observations).

Bull/Low-Risk observations make up the large majority of the sample (83.7%), visually illustrating the class imbalance discussed in Section XI.4.

Fig. 10. Rolling 252-day frequency of Bear/High-Risk regime observations.

This figure shows how the proportion of Bear-regime days within each trailing one-year window varies over time, spiking sharply around major market stress episodes rather than remaining constant, underscoring that regime frequency is itself time-varying.

The regime strategy achieved its intended objective of reducing drawdown, cutting maximum drawdown from −33.92% under buy-and-hold to −28.20%, and reducing annualised volatility from 17.20% to 11.56%. However, it did so at a substantial cost to compound return: CAGR fell from 12.68% to 5.80%, and the Sharpe ratio fell from 0.780 to 0.546, meaning the strategy delivered materially poorer risk-adjusted returns despite its lower absolute risk. Table 11 (yearly performance) confirms that this shortfall was not concentrated in one or two anomalous years but recurred across most calendar years in the sample, with the strategy trailing the benchmark in the majority of years, most notably during strong recovery years such as 2019, 2020, 2021, and 2023.

Table 11. Annual strategy vs. benchmark returns and average strategy exposure, 2010–2025.

Year	Strategy Return	Benchmark Return	Avg. Exposure
2010	6.05%	6.92%	98.1%
2011	−7.62%	0.00%	64.3%
2012	5.18%	13.41%	90.8%
2013	29.60%	29.60%	100.0%
2014	9.07%	11.39%	96.4%
2015	−7.42%	−0.73%	84.1%
2016	7.39%	9.54%	84.9%
2017	19.42%	19.42%	100.0%
2018	−16.74%	−6.24%	78.1%
2019	8.83%	28.88%	90.9%
2020	9.36%	22.12%	70.1%
2021	22.77%	26.89%	98.4%
2022	−21.08%	−19.44%	28.3%
2023	4.35%	24.23%	84.8%
2024	23.55%	23.31%	98.0%
2025	9.92%	16.39%	83.2%

Note: 2010 and 2025 are partial calendar years given the sample start (18 Oct 2010) and end (31 Dec 2025) dates.

Table 12 offers a plausible economic explanation for this pattern. Mean forward returns following a Bear/High-Risk observation are, at every horizon tested, larger than mean forward returns following a Bull/Low-Risk observation: at the 20-day horizon, the average forward return following Bear is 2.50%, versus 0.72% following Bull, and the probability of a positive 20-day forward return is likewise higher following Bear (69.8%) than following Bull (67.6%). This is consistent with a mean-reversion pattern in which the sharpest subsequent rebounds tend to follow the deepest, most volatile drawdowns — precisely the periods the tactical strategy is designed to sit out. A strategy that reduces exposure during high-volatility, high-drawdown periods will, on this evidence, tend to miss a disproportionate share of the market's strongest recovery days, which is the most direct explanation for why lower realised risk did not translate into a higher Sharpe ratio in this sample.

Fig. 11. Annual returns, regime strategy vs buy-and-hold, by calendar year.

The benchmark outperforms the strategy in most calendar years, particularly strong recovery years, consistent with the year-by-year results in Table 11.

Fig. 12. Mean forward returns by regime and forecast horizon (1, 5, 20 days).

Forward returns following Bear/High-Risk observations exceed those following Bull/Low-Risk observations at every horizon shown, illustrating the mean-reversion pattern discussed in Section IX and Section X.

Table 12. Mean forward returns and probability of a positive forward return by regime and horizon.

Regime	Horizon	Mean Fwd. Return	% Positive
Bull / Low-Risk	1-day	0.037%	54.4%
Bull / Low-Risk	5-day	0.184%	60.5%
Bull / Low-Risk	20-day	0.720%	67.6%
Bear / High-Risk	1-day	0.128%	55.8%
Bear / High-Risk	5-day	0.633%	61.7%
Bear / High-Risk	20-day	2.504%	69.8%

X. DISCUSSION

Put together, these results tell a mixed story rather than a straightforwardly positive one for machine-learning-based regime analysis in US equities. Regime clustering clearly works, in the sense that it recovers a partition of the data that is statistically distinct (Section VI.D) and economically coherent across five independent dimensions — return, volatility, implied risk, trend, and drawdown (Section VI.C) — without ever being told in advance what “bull” or “bear” is supposed to mean. The regimes are also highly persistent (Section VI.F), and that persistence is itself a meaningful finding: whatever risk state the market is currently in tends to stick around, at least in the short run, rather than flipping unpredictably from one day to the next.

However, persistence is precisely what makes next-day forecasting difficult to improve upon with supervised learning. Because a simple “tomorrow equals today” rule already captures the dominant persistence structure in the data, the marginal information that a supervised classifier can add is confined to the comparatively rare transition days, and none of the three models tested were able to exploit this margin sufficiently to beat the baseline overall (Section VII). This is consistent with the broader empirical asset-pricing literature's finding that machine learning gains over simple benchmarks in financial time series tend to be modest and are easily overstated if a naive baseline is not reported ([9]).

The lower Sharpe ratio and CAGR of the regime-conditioned strategy relative to buy-and-hold, despite its lower drawdown and volatility, illustrates a trade-off long recognised in the tactical-allocation literature ([7]): reducing exposure during identified high-risk periods lowers realised risk but also forgoes part of the return generated during recoveries from those same periods, and the forward-return evidence in Table 12 indicates this cost was material in the present sample. For an investor whose objective function places a high weight on drawdown avoidance — for example, due to leverage constraints, liquidity needs, or loss-aversion preferences — the strategy may nonetheless be attractive despite its lower Sharpe ratio; for an investor solely maximising long-run compound return, the evidence here does not support switching out of a buy-and-hold approach on the basis of this particular regime signal and cost structure.

For researchers, the paper's central methodological implication is that unsupervised regime identification, supervised regime forecasting, and economic/investment usefulness are three distinct empirical questions that should be evaluated and reported separately. A study that reports only in-sample cluster

characteristics, or only same-day classification accuracy, or only strategy backtest returns in isolation, risks conveying a more favourable impression of “ML predicting the market” than the underlying evidence supports.

XI. ROBUSTNESS AND LIMITATIONS

A. *Ex-Post Regime Identification*

The K-Means and GMM clustering was fit using the full standardisation and PCA transformation, and the Bull/Bear labels themselves were assigned using each cluster's full-sample mean return (Section V.F). This means the regime labels are ex-post descriptive classifications of the historical sample, not signals that could have been generated by an observer standing at any single point in time without knowledge of subsequent (or, in the case of the label direction, contemporaneous full-cluster) return outcomes. The out-of-sample forecasting exercise in Section VII partially addresses this by using only lagged features to predict a future label, but the label definitions themselves, and the standardisation/PCA/clustering parameters, were estimated using the training-period data as a whole rather than in a strict walk-forward, expanding-window manner.

B. *Look-Ahead Bias in the Strategy*

The investment strategy explicitly lags the regime signal by one trading day so that a given day's realised return is never used to determine that same day's investment exposure (Section V.J). This mitigates, but does not eliminate, look-ahead concerns, since the regime classification model that generated the signal was itself fit using full-sample cluster centroids and label definitions rather than being re-estimated in a purely sequential, expanding-window fashion.

C. *The Persistence Benchmark Is a High Bar*

The strength of the persistence baseline (93.55% accuracy) reflects genuine, high empirical persistence in the underlying regimes (Section VI.F) rather than a weak or poorly chosen benchmark. Any future claim of forecasting improvement in this setting should continue to be benchmarked against persistence rather than against uninformative baselines such as random guessing or the majority class alone.

D. *Class Imbalance*

The Bull/Low-Risk regime represents 83.7% of observations. This imbalance inflates the accuracy of any rule that favours the majority class, including both the persistence baseline and, to some extent, the supervised classifiers, whose precision and recall for the Bull class exceed those achievable for the comparatively rare Bear class. Reported accuracy figures should therefore be interpreted alongside this imbalance rather than as a class-neutral measure of forecasting skill.

E. *Model Limitations*

Only three supervised architectures (Logistic Regression, Random Forest, Gradient Boosting) were tested, using only the

sixteen lagged features listed in Section IV and no additional engineered features (e.g., interaction terms, longer lags, or regime-duration-aware features). It is possible that a different feature set, a probabilistic (GMM-based) target, or a sequence-aware model (e.g., a hidden Markov model estimated directly on transition dynamics) would perform differently; this was not tested in the present study and should not be assumed.

F. *Transaction Costs*

A flat 10 basis point cost per full position change was assumed. Actual costs vary by instrument, execution venue, and market conditions (costs are typically higher during the high-volatility periods the strategy trades around), so the reported net strategy performance should be treated as illustrative rather than as a precise estimate of a specific investor's achievable return.

G. *Data Limitations*

The analysis relies on a single benchmark index (the S&P 500) and a limited set of daily market, volatility, and macro indicators. No firm-level, sector, or cross-asset data were used, and the sample, while spanning over fifteen years, contains a single dominant equity bull market interrupted by a comparatively small number of drawdown episodes, which may limit the generalisability of the regime characteristics to other historical periods with different macroeconomic conditions.

H. *External Validity*

Because the sample is drawn exclusively from US large-cap equities over 2010–2025, the findings should not be assumed to generalise directly to other asset classes, other national equity markets, or other historical eras without independent testing.

I. *Future Research*

Several extensions would strengthen the robustness of these findings and are recommended for future work rather than claimed here: expanding-window, walk-forward re-estimation of the clustering and forecasting pipeline (so that no future information enters any single day's classification); explicit hidden Markov model estimation of the transition dynamics as a parametric alternative to K-Means/GMM; rolling or online clustering that allows the regime definitions themselves to evolve; application of the same pipeline to international equity indices and to other asset classes (fixed income, commodities, currencies); incorporation of additional macroeconomic variables (e.g., credit spreads, yield-curve slope, ISM survey data); sensitivity analysis of the tactical strategy's results to alternative transaction-cost assumptions and to partial (rather than binary 0%/100%) position sizing; and regime-dependent portfolio optimisation that allocates across multiple assets conditional on regime rather than adjusting a single equity

exposure. Section XI.J provides an initial, coarse test of one item on this list — sensitivity of the regime solution to the feature set — and finds the main specification's exclusion of non-stationary raw price and volume series to be well justified; a natural next step would extend that test with a walk-forward re-estimation of the widened specification to establish whether its forecasting collapse is a property of the feature set alone or is compounded by the full-sample estimation limitation noted in Section XI.A.

J. Robustness Check: Widening the Feature Set and Regime Count

As a robustness check on the feature-exclusion decision described in Section IV, the full clustering and forecasting pipeline was re-run on a wider specification of the same underlying dataset: twenty-nine standardised features rather than sixteen, restoring the raw OHLC price levels, raw trading volume, the 20-/50-/200-day moving averages, and the running maximum price that the main specification deliberately excludes for being non-stationary (Section IV). Every other step of the pipeline — PCA to 90% cumulative explained variance, K-Means with silhouette, Calinski-Harabasz, and Davies-Bouldin model selection, GMM validation by BIC, an identical chronological 70/30 split, and the same set of forecasting models — was left unchanged, so that any difference in outcome can be attributed to the feature set itself rather than to a change in methodology. This specification increased the observation count slightly ($n = 3,823$, unchanged) and PCA now retains nine components to reach 90.3% cumulative variance, versus eight components and 92.1% in the main specification.

The result is a negative one, and it argues for, rather than against, the original exclusion. Cluster quality is weaker and the three internal validation metrics no longer agree with one another. Table 13 reports the K-Means diagnostics for $K = 2$ through 6 under the widened feature set: the best silhouette score achieved anywhere in this range is 0.324 (at $K = 4$), well below the 0.466 that $K = 2$ achieved decisively in the main specification (Table 3), and each of the three metrics now points to a different value of K . With no metric agreement, the selection procedure's tie-break rule resolves to $K = 4$ by list order rather than by genuine three-metric consensus. The Gaussian Mixture Model tells a similar story: BIC declines monotonically out to the upper bound tested (six components) instead of settling on an interior minimum, in contrast to the two-component, full-covariance GMM that corroborated the main K-Means solution in Section V.E.

The resulting four regimes are also less economically coherent than the original two. The regime auto-labelled “Bullish” by

mean return (mean VIX 29.12) carries a higher average VIX than the regime auto-labelled “Bearish / High Volatility” (mean VIX 28.01) — the two labels are, on this measure, essentially indistinguishable and arguably reversed. Historical-episode classification compounds the problem: 59.4% of trading days during the 2022 bear market are assigned to the “Bullish” regime, and no single regime accounts for a majority of the COVID crash period (the largest share, 40.9%, again falls to the “Bullish” regime). By contrast, the main two-regime specification places the COVID period and the 2022 drawdown predominantly in Bear/High-Risk, consistent with its economic characterisation in Section VI.C.

Table 13. K-Means and GMM validation diagnostics under the widened (29-feature) specification, $K = 2$ through 6.

K	Silhouette	Calinski-Harabasz	Davies-Bouldin	GMM BIC
2	0.317	1,405.30	1.466	112,939.99
3	0.323	1,251.47	1.452	109,045.46
4	0.324	996.45	1.590	106,263.90
5	0.191	884.53	1.711	104,451.95
6	0.182	815.76	1.755	102,097.60

Note: Under the main 16-feature specification, silhouette, Calinski-Harabasz, and Davies-Bouldin all favoured $K = 2$ (Table 3). Under the widened specification the three metrics disagree — silhouette favours $K = 4$, Calinski-Harabasz favours $K = 2$, and Davies-Bouldin favours $K = 3$ — and the reported $K = 4$ solution is the pipeline's default tie-break rather than a genuine three-metric consensus. GMM BIC declines monotonically to the upper bound of the range tested ($K = 6$) rather than reaching an interior minimum, versus the two-component GMM that corroborated the main specification (Section V.E). Source: Replication Data Archive, sheets Supplementary Appendix, Table S13 and Supplementary Appendix, Table S13.

The forecasting exercise deteriorates even more sharply than the clustering, and the mechanism is identifiable rather than mysterious. Because the widened feature set reintroduces non-stationary raw price levels, the regime definitions drift with the price level itself: one regime accounts for only 5.2% of the chronological training window but 79.7% of the test window, since the test period (the most recent, highest-price-level trading days) is dominated by a cluster that barely existed earlier in the sample. This is precisely the look-ahead-adjacent failure mode that the exclusion of raw price levels in Section IV was designed to prevent, now demonstrated empirically rather than assumed. Table 14 shows the consequence: Random Forest's balanced accuracy is exactly 33.33% at all three

horizons, indicating the model has collapsed to predicting a single class regardless of input; Logistic Regression falls to a literal 0% accuracy at the 5-day horizon; and Gradient Boosting swings from 15.3% accuracy at the 1-day horizon to 78.3% at the 5-day horizon, a horizon-to-horizon instability not present anywhere in the main specification's results (Table 8).

Table 14. Out-of-sample regime forecasting performance under the widened (29-feature, four-regime) specification, chronological 70/30 split, three horizons.

Model	Horizon	Accuracy	Balanced Accuracy	Macro F1
Persistence Baseline	1-day	91.28%	68.28%	68.28%
Logistic Regression	1-day	71.67%	29.98%	23.10%
Random Forest	1-day	79.69%	33.33%	29.56%
Gradient Boosting	1-day	15.26%	27.11%	12.57%
XGBoost	1-day	80.56%	35.69%	34.27%
Persistence Baseline	5-day	87.78%	62.63%	62.63%
Logistic Regression	5-day	0.00%	0.00%	0.00%
Random Forest	5-day	79.67%	33.33%	29.56%
Gradient Boosting	5-day	78.27%	39.57%	42.00%
XGBoost	5-day	78.45%	32.82%	29.31%
Persistence Baseline	20-day	78.53%	49.22%	49.22%
Logistic Regression	20-day	65.91%	27.61%	21.52%
Random Forest	20-day	79.58%	33.33%	29.54%
Gradient Boosting	20-day	73.36%	32.37%	29.73%
XGBoost	20-day	79.40%	33.26%	29.51%

Note: Random Forest's balanced accuracy of exactly 33.33% at every horizon indicates the model degenerates to predicting a single class regardless of input. Logistic Regression collapses to 0% accuracy at the 5-day horizon. Persistence remains the strongest single model by balanced accuracy at every horizon, as in the main specification (Table 8), but at markedly lower absolute levels than the two-regime case.

Source: Replication Data Archive, sheet Supplementary Appendix, Table S14.

Taken together, this robustness check does not overturn any conclusion drawn from the main two-regime specification; it instead provides an empirical, rather than purely a priori, justification for the feature-exclusion decision in Section IV. Restoring non-stationary raw price and volume series does not reveal additional regime structure the main specification missed — it degrades cluster validity, weakens economic coherence, and, through a severe temporal distribution shift, destroys the very forecasting signal the rest of the paper depends on. This is offered as a cautionary result for future work extending the feature set (Section XI.I): more features are not automatically better features in a regime-clustering context, and any expansion of the feature set should be checked for stationarity and for train/test distributional stability before being adopted.

XII. CONCLUSION

This paper set out to answer four research questions concerning US equity market regimes. Regarding RQ1, the evidence strongly supports the existence of statistically and economically distinct regimes: an unsupervised pipeline combining PCA, K-Means, and GMM validation identifies a two-regime solution that differs significantly, with large effect sizes, across return, volatility, VIX, momentum, and drawdown. Regarding RQ2, the identified regimes exhibit strong temporal persistence, particularly the Bull/Low-Risk regime (96.5% day-to-day continuation probability), though regime duration is highly right-skewed and the “typical” episode is considerably shorter than the sample mean would suggest. Regarding RQ3, the evidence does not support the hypothesis that supervised machine learning forecasts next-day regime membership better than a naive persistence rule; the persistence baseline outperformed Logistic Regression, Random Forest, and Gradient Boosting on out-of-sample accuracy. Regarding RQ4, a regime-conditioned tactical strategy reduced both volatility and maximum drawdown relative to buy-and-hold but underperformed on compound annual growth rate and Sharpe ratio, a shortfall plausibly linked to systematically stronger forward returns following high-volatility, drawdown-heavy (Bear) observations than following calmer (Bull) observations.

The overarching conclusion is that market regimes can be identified in a statistically and economically meaningful manner, but that identifying regimes after the fact is not equivalent to reliably forecasting regime transitions before they occur, nor to generating superior risk-adjusted investment returns once a regime signal is translated into an implementable, cost-adjusted strategy. This distinction — between descriptive classification, predictive forecasting, and economic usefulness — is the paper’s central and most defensible finding, and it is offered as a caution against interpreting strong in-sample or same-day regime-classification results as evidence of tradable predictive skill.

The robustness check in Section XI.J adds a further, more specific caution. Widening the feature set to include non-stationary raw price and volume series does not uncover richer regime structure; it produces a weaker, less internally consistent clustering solution and, through a severe temporal train/test distribution shift, a near-total collapse in out-of-sample forecasting performance. Read alongside the main results, this is best understood not as a competing finding but as an empirical stress-test of a modelling choice made in Section IV — one that the choice survives.

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XIII. APPENDIX

A. Data Quality Summary

The cleaned dataset contains 3,823 observations across 33 columns with zero missing values in any column after the cleaning steps described in Section V.A (infinite-value removal, duplicate-date removal). The full column-by-column data-quality table is available in the Supplementary Appendix, Table S1.

B. Full PCA Loadings (Selected Variables, PC1–PC3)

Table A1. Selected PCA loadings for the first three principal components (full 16×8 loading matrix available in Supplementary Appendix, Table S3).

Feature	PC1	PC2	PC3
Return	0.137	0.473	−0.223
Volatility_20D	−0.325	0.233	0.204
Momentum_60D	0.311	−0.100	0.116
Distance_MA50	0.347	0.023	0.338
Distance_MA200	0.341	−0.116	0.016
Drawdown	0.379	−0.149	−0.141
VIX	−0.338	0.107	0.213
VIX_Change	−0.109	−0.430	0.223
Treasury_20D_Change	0.223	0.000	0.228
Oil_Return	0.061	0.198	0.034

C. Reproducibility Note

All figures, tables, and numerical results reported in this paper were extracted directly from the project Replication Data Archive and its Supplementary Appendix. No values were estimated, simulated, or approximated for the purposes of this paper.